3.4 Measures to reduce the hazard

When avoidance is not an option, other mitigation measures must be considered. Mitigation measures which aim to reduce the hazard must reduce the probability of triggering of the landslide(s) which the specific measure is intended to address. This type of mitigation measures are sometimes referred to as "stabilization".

The stability of any hillslope is decided by a balance of forces between stresses driving downward movement (shear stress) and stresses resisting to movement (shear strength) at the bottom interface between the landslide mass and the stable material, with gravity as the primary driving force (Lu and Godt, 2013). Thus, factors (or forcings) that affect the shear strength (e.g., weathering, vegetation, past movement, variation in pore-water pressure or changes in hydrologic conditions/properties) or the shear stress acting on the hillslope (e.g., mass redistribution, debuttressing at the toe, transient loading from earthquakes) are potential causes to landslide failure and/or acceleration (Sidle and Ochiai, 2006; Lu and Godt, 2013; Lacroix et al., 2020).

This mechanical balance between driving and resisting forces is mediated by the presence of water in the slope and quantified using the concept of effective stress (Lu and Godt, 2013). Water infiltration will lead to the build-up of pore-water pressure within the slope (generally depicted by the transient development of a perched water table), which will typically reduce the shear strength of the slope material (Sidle and Ochiai, 2006) and increase shear stress due to the weight of water (Lu and Godt, 2013). Besides, interactions between forcings – such as the combination of rainfalls with earthquake(s) – may affect the slope stability further than each forcing individually, illustrating the complexity of landslides mechanisms (Lacroix et al., 2020). It is to be noted that, when considering factors conditioning the hillslope stability, it is important to account that the forcings triggering initial failure (i.e., landslide occurrence) may be different from forcings controlling later motion (if any).

Independently of the causative processes and the complexity of the specific geosystem under consideration the factors which determine the triggering of movements are:

a) decrease in shear strength
b) increase in shear stress

The ratio between shear strength and shear stress forms the factor of safety (FS), i.e. an easy approach to discuss what drives the stability of a slope. This ratio may change over time in response to environmental changing conditions (Figure 12 and Figure in Appendix 1.2).